3.3 Vectors in the Plane

Name _____ Period ____ Date _____

Pre-Calculus 3.3 (30-minute plan) · Student sheet

I can…

- write the component form of a vector and find its magnitude.
- add, subtract, and multiply vectors by scalars, and find unit vectors.
- use direction angles to write vectors and solve real-life problems.

Vocabulary · fill in the missing word

Vector — a quantity with _____ and direction, shown as a directed line segment PQ (initial point P, terminal point Q)

Component form — $v = $ _____, $v_2\rangle = \langle q_1 - p_1, q_2 - p_2\rangle$

Magnitude — $\|v\| =$ _____

Unit vector — a vector with _____ 1: $v/\|v\|$

Standard unit vectors — $i =$ _____, $0\rangle$ and $j = \langle 0, 1\rangle$

Direction angle — the angle θ from the _____ x-axis: $v = \|v\|\langle \cos\theta, \sin\theta\rangle$

Part 1 · Notes: fill in the blanks

The component form of the vector from $P(p_1, p_2)$ to $Q(q_1, q_2)$ is _____.

Magnitude: $\|v\| =$ _____

$u + v =$ _____ $ku =$ _____

A unit vector in the direction of v is _____.

Direction angle: $v = \|v\|\langle$_____, _____$\rangle$

Part 2 · Examples (work them with the class)

Example 1

Find the component form and magnitude of the vector with initial point $P(4, -7)$ and terminal point $Q(-1, 5)$.

Example 2

Let $u = \langle -2, 5\rangle$ and $v = \langle 3, 4\rangle$. Find each vector.

a. $2v$ b. $u + v$ c. $v - u$ d. $2u + 3v$

Example 3

Find the magnitude and direction angle of each vector.

a. $u = 3i + 3j$ b. $v = 3i - 4j$

Part 3 · Partner practice

Partner practice

Let $u = \langle 4, -1 \rangle$ and $v = \langle -2, 3 \rangle$. Find $3u - v$ and its magnitude.